

NEET	2024	2023	2022	2021
Animal Kingdom	1) Eg of fish 2) Invertebrates 3) Features of Nonchordata	1) Features of mammals (Mixed) 2) Features of Chordata 3) Bases of classification 4) Invertebrate (Mixed)	1) Taxonomic Classification 11) Arthropoda 111) Features of Classification	1) Ecnthoon 2) Invertebra 3) Invertebra 4) Birds 5) Arthropoda 6) Invertebra
Structural Organisation	1) Cockroach 2) Cockroach 3) Epithelial Tissue	1) Cockroach-E 2) Cockroach-Stru 3) Tissue (Mixed) 4) Tissue-Connective	1) Tissue (Mixed) 2) Connective Tissue 3) Cockroach	1) Type of junction 2) Cockroach
Breathing	1) Pulmonary Volume & Capacities 2) O_2 -Hb-Dissoc curve	1) Pulmonary Volume & Capacitance	1) Function of Conducting Part 2) Transport of O_2	1) Altitude Sickness 2) Pp of O_2 & CO_2 3) O_2 -Hb-Dissociation Curve
Body fluid & Circulation	1) Heart = Conducting Pathways 11) Heart-ECG 111) Blood Group	1) Types of WBC 11) Heart-ECG 111) RBC = Hb	Heart = function Blood = Clothing	1) Blood Group 11) Blood Clothing
Excretory product & their Elimination	1) Nephron = function 11) Juxta-Medullary Nephron	1) Hormonal Regulation 11) Juxta Medullary Nephron	1) Types of Nitrogenous Waste in Animal (Mixed)	1) Juxta Glomerular cell - Erythropoietin
Locomotion	1) Type of Muscles 2) Joint 3) Disorder	1) Skeletal Muscles 2) Joint	Joint Disorder (Muscle) Disorder = Osteoporosis	1) Joints 2) Disorder 3) Muscles 4) Smooth Muscle

NEET	2024	2023	2022	2021
Neural Control & Coordination	1) Brain Structure 2) Brain function	Limbic System function	1) Types of Synapses	
Chemical Coordination	1) Steroid hormone 2) Action-Mechanism 3) Disorder	1) Thyroid Hormone 2) function of Hormone (Mix)	1) Function of Parathyroid Hormones	
Human Reproduction	1) External Genitalia 2) Lactation 3) Femal Repro. Structure 4) Spermatogenesis	1) Menstrual cycle 2) Implantation 3) Female - Repro Structure	1) Oogenesis 2) Spermatogenesis 3) Oogen-Sper (mixed)	1) Parturition 2) " 3) fertilization
Reproductive Health	IUD Contraceptive Method	1) Amniocentesis 2) STD 3) Contra Methods	IUD Contra - Method	IUD Contra method
Human Health & Disease	1) Human Disease 2) " 3) Drugs 4) Lymphoid organ	1) HIV-Lifecycle 2) Human Disease 3) Drugs	1) Immune Response 2) Human Disease 3) Autoimmunity	1) Human Disease 2) Viral Disease
Evolution -	1) Human Exa 2) Hardy Weinberg equation 3) Homologous 4) Exa & Animals	1) Adaptive Radiation	1) Homologous Org. 2) Natural Selection	1) Adaptive Radiation 2) " 3) Genetic Drift (founder effect)
Bio tech principle	1) Restriction Enzyme (HMD-III) 2) Cloning Vectors 3) Bio-reactors 4) " 5) PBR 322	1) Restriction Enzyme [Cloning Vectors] 2) RDT - Steps 3) Gene Gun 4) Isolation of DNA 5) Gel Electrophoresis	Cloning Vector Restriction Endo Restriction Enzymes Cloning Vector Gel Electrophoresis PCR & Selectable marker	1) Restriction Enzyme 2) PBR - 322 3) PCR 4) PCR 5) Isolation of DNA

S.No.	Zoology Low Weilage	Botany High Weilage	
Chapter	Biotechnology P & P.	Molecular Basis of Inheritance	
	• Principle of Biotech. • Tools of Recombinant DNA Technology • Process of Recombinant • Restriction Enzymes • Cloning Vectors • Bio - Reactors • PBR 3²² • RDT-Step • Gene Gun • Isolation of DNA • Gel Selection Electron • Restriction endonuclease • PCR & selectable marker • Isolation of DNA	78 • Structure of Nucleic Acid. • Discovery of DNA • Watson & Crick Model • Packaging of DNA • Search of Genetic Material • Experiment & Process • Transcription Process • Type of RNA • Genetic Code • Translation • Gene Expression • Human Genome Project • DNA fingerprint.	
Chapter	Biotechnology & Appl. Biotechnological Application in Agriculture Biotechnological Application in Medicine Transgenic Animals Ethical Issue	Principle of Inheritance → Intro - Gen allele chromosome → Haploide, diploid → Mendelian principle → Hybridization experiment Result law & cross Gene interaction - • Incomplete Dominance • Multiple Allele • Polygenic inheritance. • Chromosomal theory, linkage. • Sex determination. • Mutation pedigree. • Genetic Disorder.	Basic → Gene → alleles → Chromosome → Chromatin → genome, genotype → gene pool phenotype • Concept of character → Concept of locus.

Chapter	Human Reproduction	Organism & Population
	• External Genitalia • Lactation • Female Reproductive Organ (Structure) • Spermatogenesis • Menstrual cycle • Implantation • Oogenesis • Oogenesis + sperm (mix) • Parturition • Fertilization • Types of Sex Organ • Male & female Repro • Testis & Ovary • Mechanism of ejaculation & excretion • Male & female sex Accessory Gland. • Mammary Gland & Vulva • Embryonic development • Placenta • Gestation • Parturition	• Birth Rate • Sex Ratio • Expanding • Declining • Parthenium • Bacteria tiger fish • Pop. Natality • Immigration • Exponential Model • Biotic potential • Equation • Asymptote • Death Rate • Age Distribution • Stabilizing • population D. • Population Growth • Mortality • Emigration • Base of Natural Logarithm • logistic Model • Carrying Capacity • Population Interaction • Symbol • Predation • Prudent • Avoidance of predation • Ammensalism • Example • Role of Predation • Competition • Abingdon tortoise • Competitive Release • Graze Exclusion • Resource partitioning • Parasitism • Ectoparasite • Coevolution • Commensalism • Cattle egret • Orchid • Mutualism • Lichen, Mycorrhizae • Endo. • Brood Parasitism • Clown fish • Wasp • Oryctolagus • pollinator

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n.	Ecosystem	Reproductive Health	
	• Introduction • Productivity • Decomposition • Energy flow • Ecological pyramid (mixed)	• IUD Contraceptive • Amniocentesis • STD • population explosion • MTP + STD + Infertility	
Ch.	Biodiversity	Human Health & Disease	
	Biodiversity Level of Biodiversity How much Pattern Significant Loss Reason Conservation Scientist Name. Bio Diversity	→ Human Disease → HIV - life cycle → Immune Response → Human Disease & Viral Disease (Mixed Question) → Drugs. → Lymphoid Organ → HIV + Autoimmunity → Immunity → Acquired Immunity → Active & Passive Immunity → Allergies → Vaccination & Immunisation → Immune System → Protozoan, Cancer → AIDS	

CH.	Microbes & Human Welfare	Evolution	Structural Organisation in Animals
	Exple Example of Microbes Household plant Industrial plant Sewage treatment plant Biogas plant Bio reactor + Biofertiliser	• Human Era • Hardy Weinberg equation • Homologous • Adaptive Radiation • Natural selection • Genetic Drift (founder effect) • Darwinism • Mutation theory • Hardy Weinberg Principle • Brief Account of evolution • Evidence in favour evolution	• Cockroach • Epithelial Tissue • Tissue Conn^{tion} • Connective tissue • Type of junction • Probability • Type of Cartilage & Bone • Exoskeleton • Frog
CH.	Plant kingdom	Breathing & exchange of Gases	
	• Algae Bryophyte • Pteridophyte • Bryophyte • Gymnosperm • Angiosperm	Pulmonary Volume & cap. O₂ Hb - Dissociation curve function of conducting Transport of O₂ Altitude sickness PP of O₂ & CO₂ Machanism of Breathing Transport of Gases Regulation of Breathing & Disorde	

Ch.	Higher Photosynthesis in Plants	Locomotion & Movement
	Experiment location of photosynthesis Type of pigment <u>Light Reaction</u> Dark Reaction Chemiosmotic Hypothesis Photorespiration factor Affecting Rate of Photosynthesis	- Types of Muscles → Joints → Disorder → Skeletal Muscles → Joint Disorder → Disorder - Osteoporosis → Muscles contractions → Smooth Muscles Structure of contractile Protein Difference  Human Skeletal system
CH -	Respiration in plant Need of Respiration Autotrophic & Hetero trophic cell Respiratory Substrate Plant Breath Glycolysis Anaerobic Ethanol Lactic Acid Fermentation Aerobic Respiration Link Reaction Krebs cycle Σ TS Amphibolic + RCE	Neural control & Coordination Brain Structure Brain function Limbic System Types of Synapses Types of Nephrons Conduction Nerve Impulse CNS - Brain

CH - Morphology in flowering	Body fluids & circulation
Inflorescence Flower Fruit family 3-4 Q Root Stem leaf 1 Q	Heart - Conducting Pathways Heart - ECG Blood Group Types of W.B.C R.B.C - H.b Heart - Function Blood - clotting Blood - Group - circulating pathways - Breathing Heart Rate Heart → Heart sound - Cardiac cycle - Disorders → ECG +
Anatomy of flowering	Excretory product & el.
→ Dicot Root Monocot Root Dicot Stem Monocot Stem Meristematic Tissue Simple Permanent tissue Complex permanent Tissue • Tissue System → Root > system	→ Nephron Function → Juxta Medullary Nephron Hormonal Regulation → Type of Nitrogenous Waste in Animals (Mixed) → Juxta medullary + Glomerulus cell → Erythropoietin Urine formation + Function of Tubules + Counter current Mechanism → Regulation of Kidneys → Disorder of Excretory system

CH.	Plant Growth & Development	Chemical Coordination & Integration.	7
	Growth Phase of Growth. Arithmetic & Geometric Differentiation Development Plant Growth Regulation Auxin Cytokinin Ethylene G.A. <u>Deleted</u> - Photo Respiration Vernalization Dormancy	- Steroid Hormone - Action Mechanism - Disorder → Thyroid Hormone. → Function of Hormone. Function of Polthyracoid Hormone.	
CH.	Biological Classification	Biomolecules	
	Monera Virus ↑ Protist Viroid ↑ Fungi Prions ↓ Special focus on. Monera Nutrition Fungi Example Spore formation - Scientist Virus Viroid Classification of System.	Elemental Analysis Chemical Analysis - Primary Metabolite. Secondary Metabolite → Amino Acid. → Protein → Nucleic Acid. → Enzyme → Lip. → Carbohydrate.	

CH.	Biological classification	Biomolecules
	Monera Virus ↑ Protist Viroid Fungi Prions ↓ Special focus on. Monera Nutrition Fungi Examples Spore formation - Scientist Virus Viroid Classification of System.	Elemental Analysis Chemical Analysis - Primary Metabolite. Secondary Metabolite → Amino Acid. → Protein → Nucleic Acid. → Enzyme → Lip. → Carbohydrate.

CH.	Unit Cell Cycle Division	Cell cycle	Animal Kingdom
	Introduction cell theory. Basic of cell Prokaryotic cell Eukaryotic cell. Vacuole plastids. Cytoskeletal. Element ① Cell Membrane ② cell wall ③ Endoplasmic Reticulum. ④ Endomembrane System ⑤ Golgi bodies. ⑥ Lysosome. ⑦ Mitochondria. ⑧ Ribosome ⑨ Nucleus. ⑩ Microbodies	Basics of cell cycle • Mitosis. • Meiosis. (Understand)	Eg. of Fish Invertebrates. Feature of Nonchordata Features of mammals Features of chordata Base of classification Invertebrates (mixed) Taxonomic classification → Arthropoda features of classification → Earthworm Birds → Invertebrates.
Ch.	Living World →	Sexual Reproduction in flowering plants.	
	→ Features of living organisms Taxonomy Nomenclature Taxonomic category	Structure of Anther & Gametogenesis. Structure of pistil & Gametogenesis. - Pollination outbreeding → Pollen pistil interaction → Double Fertilization	→ Embryo development. → Seed development. → fruit development → Apomixis & Polyembryonic development.

Biology Important Topics Checklist (☺_☺)

S.No	Chapter	Topics / Checklist Items	Status
1	Living World	• Table Data • Examples (Canidae, Felidae)	☐
2	Biological Classification	• Examples - Fungi • Virus - Scientist Contributions	☐
3	Plant Kingdom	• First page concepts • Pteridophyte Examples • Economic importance	☐
4	Animal Kingdom	• Examples (Specific focus)	☐
5	Morphology of Flowering Plants	• Androecium • Gynoecium • Placentation • Family characteristics (Examples likely implied)	☐
6	Anatomy of Flowering Plants	• T.S comparison (Root, Stem, Leaf - Dicot/Monocot implied)	☐
7	Structural Organisation in Animals	• Cockroach • Frog	☐
8	Cell: The Unit of Life	• Nucleus • Cytoskeleton	☐
9	Cell Cycle and Cell Division	• Prophase-I (Meiosis)	☐
10	Biomolecules	• Tables (e.g., composition) • Bonds in Carbohydrates • Nomenclature in Enzymes	☐
11	Photosynthesis in Higher Plants	• Scientist Contributions • ETS (PSI & PSII) • C4 Cycle	☐
12	Respiration in Plants	• ETS (Electron Transport System) • Enzyme names (involved in pathways)	☐
13	Plant Growth and Development	• Auxin Role • Ethylene Role	☐
14	Breathing and Exchange of Gases	• PO₂ (Partial pressure O₂) • PCO₂ (Partial pressure CO₂) • RV (Residual Volume) • ERV (Expiratory Reserve Volume) - values • Neural control	☐
15	Body Fluids and Circulation	• ECG • Disorders • WBC - age related changes (if applicable)	☐
16	Excretory Products and	• RAAS (Renin-Angiotensin-Aldosterone System) • Secretion & Absorption (in nephron)	☐

Ecosystem		
1.	Basic structure	- Stratification Component
2.	Productivity	- GPP, NPP.
3)	Energy flow	- Food chain
	Ecological Pyramid	- Energy Biomass
4.	Productivity	Primary productivity GPP NPP Comparison Numerical Secondary Productivity
5	Decomposition	litter fragmentation leaching Catabolism Mineralization Humus factor
6	Energy flow	PAR Food chain DFC GFC Stanaleicrop
7	Ecological Pyramid	Foodweb → Energy Biomass Number Inverted upright Drawback

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|----|---------------------------------------|--|--------------------------|
| 19 | Chemical Coordination and Integration | <ul style="list-style-type: none"> • Adrenal gland • Testis • Ovary • Pancreas | ☐ |
|----|---------------------------------------|--|--------------------------|

Class 12th Topics

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|----|---|--|--------------------------|
| 20 | Sexual Reproduction in Flowering Plants | <ul style="list-style-type: none"> • Examples - All (Emphasis on examples throughout chapter) | ☐ |
| 21 | Human Reproduction | <ul style="list-style-type: none"> • Embryo development stages • Oogenesis • Menstrual cycle | ☐ |
| 22 | Reproductive Health | <ul style="list-style-type: none"> • MTP (Medical Termination of Pregnancy) • Contraceptives | ☐ |
| 23 | Molecular Basis of Inheritance | <ul style="list-style-type: none"> • HGP (Human Genome Project) • Types of RNA | ☐ |
| 24 | Principles of Inheritance and Variation | <ul style="list-style-type: none"> • Co-dominance • Polygenic inheritance • Sex determination • Autosomal Recessive disorders | ☐ |
| 25 | Evolution | <ul style="list-style-type: none"> • Human Evolution - data/key stages | ☐ |
| 26 | Human Health and Disease | <ul style="list-style-type: none"> • Drugs (Abuse) • Cancer | ☐ |
| 27 | Microbes in Human Welfare | <ul style="list-style-type: none"> • Bioactive molecules • STP (Sewage Treatment Plant) • Acids (produced by microbes) | ☐ |
| 28 | Biotechnology: Principles and Processes (I & II Combined) | <ul style="list-style-type: none"> • Cloning vector • Year (Important discoveries/timeline if specified) • RNAi (RNA interference) • BT plants • Autoradiography • (Order insects Transgenic - likely related to BT plants/pest control) | ☐ |
| 29 | Organisms and Populations | <ul style="list-style-type: none"> • Population interactions - Examples | ☐ |
| 30 | Ecosystem | <ul style="list-style-type: none"> • Productivity - definition/types | ☐ |
| 31 | Biodiversity and Conservation | <ul style="list-style-type: none"> • Pie chart (representing biodiversity data) • In situ conservation • Earth summit • World summit | ☐ |

Organism & Population

→ Death rate.
 Birth rate.
 Sex Ratio
 Age distribution.
 Expanding
 Stabillizing
 Declining
 Population density.
 Parthenium Bacteria
 tiger fish.
 Population Growth.
 Natality.
 Mortality.
 Immigration
 Emigration.
 Exponential modal.
 Base of Natural logarithm
 Biotic potential.
 Logistic modal
 Equation
 Carrying Capacity.
 Asymptote.
 Population interaction.
 Symbal.
 Ammensalism
 Predation
 Example.
 Prudent.
 Role of Predation
 Avoidance of Predation

Competition
 Abingdon tortoise
 Competitive Release
 Gause Exclusion
 Resource partitioning

Parasitism
 Ectoparasite.
 Endoparasite.
 Coevolution.
 Brood parasitism
 Commensalism
 Cattle egret orchid
 Mutualism
 Lichen Mycorrhiza
 Wasp
 Ophrys.

Biodiversity

Method of Conservation
 in situ -
 Hotspot
 Sacred grove.
 Biosphere Reserve
 National park -
 Ex situ -
 Earth Summit.

	Chapter	Topics (with Weightage)	Theory Complete	Class Question Complete	BMS Prime Complete	PYQs Completed	Test	Test Analysis	Revision
1	Molecular Basis of Inheritance	The DNA - 4 Packaging of DNA Helix - 4 Transcription - 3 The Lac operon - 2 Genetic Code - 2 DNA Fingerprinting – 2 Human Genome Project – 2							
2	Principles of Inheritance and Variation	Mendelian Disorders - 4 Linkage and Recombination – 3 Inheritance of One Gene - 3 Mendel's Laws of Inheritance - 2 Sex Determination - 2							
3	Biotechnology: Principles and Processes	Restriction Enzymes – 7 Amplification of Gene of Interest using PCR - 4 Separation and isolation of DNA fragments - 3 Cloning Vectors – 3							
4	Biotechnology and its Applications	Biotechnological Applications in Agriculture - 5 Molecular Diagnosis - 3 Genetically Engineered Insulin - 3 Gene Therapy – 2							
5	Human Physiology:	Complete, Pulmonary Volume							

Sno.	Chapter	Topics (with Weightage)	Theory Complete	Class Question Complete	BMS Prime Complete	PYQs Completed	Test	Test Analysis	Revision
	Breathing and Exchange of Gases	and Capacities							
6	Human Physiology: Locomotion and Movement	Disorders, Skeletal System, Joints							
7	Human Physiology: Excretory System	• Juxtaglomerular Nephron, Regulation of urine formation							
8	Human Physiology: Body Fluids and Circulation	• Blood, Electrocardiograph (ECG), Cardiac Cycle							
9	Human Physiology: Neural control and coordination	Conduction mechanism, Brain function							
10	Human Physiology: Chemical coordination and integration	Function and disorders of Hormones							
11	Cell Cycle and Cell Division	Phases of Cell Cycle - 7 Meiosis - 6							
12	Cell-The Unit of Life	Nucleus - 4 Mitochondria - 3							
13	Plant Kingdom	Algae - 6 Pteridophytes - 3 Gymnosperms - 3 Liverworts - 2							
14	Animal Kingdom	Phylum - Chordata - 5							
15	Anatomy of Flowering Plants	Complex Tissues - 2 Spring wood and autumn wood - 2 Monocotyledonous Plant - 2							
16	Structural Organisation in Animals	Cockroach - 5 Epithelial Tissue - 4							

Chapter	Topics (with Weightage)	Theory Complete	Class Question Complete	BMS Prime Complete	PYQs Completed	Test	Test Analysis	Revision
	Frog -Circulatory and Reproductive system							
	Evolution							
	What are the Evidences for Evolution? - 5							
	Hardy-Weinberg Principle – 4							
18	Plant - Growth and Development							
	Auxins - 3							
	Ethylene – 3							
	Development - 2							
	photoperiodism - 2							
19	Human Reproduction							
	Gametogenesis - 3							
	Fertilisation and Implantation – 3							
20	Human Health and Disease							
	Common Diseases in Humans - 4							
	Immunity – 5							
21	Biomolecules							
	primary and Secondary Metabolite - 3							
	How to Analyse Chemical Composition? – 2							
22	Sexual Reproduction in Flowering Plants							
	Pollination - 3							
	The Pistil, Megasporangium (ovule) and Embryo sac - 2							
	Seed - 2							
	Double Fertilisation – 2							
23	Morphology of Flowering Plants							
	Description of Some Important Families - 5							
	The Flower - 4							